

VPA ICT — IGCSE Y9S1 Weeks 10–18: Data, Effects, Security & Systems

Weeks 10 – 18: Data Types, Effects of ICT, Security & Systems Analysis

Syllabus: CIE IGCSE ICT (0417) Topics 5, 6, 8, 9 | Lesson Length: 40 minutes

Week 10 — Data Types: Logical, Text, Numeric

Syllabus: 5.1 Data Types

Keywords: Boolean 布尔型 (bù'ěr xíng), string 字符串 (zìfú chuàn), integer 整数 (zhěngshù), real/float 实数/浮点数 (shíshù/fúdiǎn shù), date/time 日期/时间 (rìqī/shíjiān), char 字符 (zìfú)

Activity 1 (15 min): Data type sorting — Given 20 data items (e.g., "123", "Hello", TRUE, 3.14, 2024-01-15), sort into correct type categories.

Activity 2 (15 min): "Choose the right type" — For each field in a database (Age, Phone, Price, IsMember), select the most appropriate data type and justify.

Activity 3 (10 min): Validation link — How does data type choice affect what validation rules are possible?

Exam Tip: For database questions, always choose the MOST SPECIFIC type. Age = INTEGER (not text). Price = REAL/CURRENCY (not integer). Phone = TEXT (not numeric — leading zeros matter!).

Week 11 — Data Structures: Arrays, Records, Files

Syllabus: 5.2–5.3 Data Structures

Keywords: array 数组 (shùzǔ), record 记录 (jìlù), file 文件 (wénjiàn), field 字段 (zìduàn), key field 关键字段 (guānjiàn zìduàn), fixed length 固定长度 (gùdìng chángdù), variable length 可变长度 (kěbiàn chángdù)

Activity 1 (15 min): Array practice — Given array of 10 temperatures, find highest, lowest, average. Trace table exercise.

Activity 2 (15 min): Record design — Design a student record with 6 fields. Identify the key field. Why is it unique?

Activity 3 (10 min): Fixed vs. variable length — What are the advantages of each? When is fixed length faster?

Exam Tip: For trace table questions, create a table with columns for each variable and work through the algorithm ONE STEP AT A TIME. Don't skip steps — examiners check each line.

Week 12 — Effects of ICT: Social & Economic

Syllabus: 6.1–6.2 Social and Economic Effects

Keywords: unemployment 失业 (shīyè), teleworking 远程办公 (yuǎnchéng bàngōng), e-commerce 电子商务 (diànzǐ shāngwù), globalization 全球化 (quánqiú huà), digital divide 数字鸿沟 (shùzì hónggōu), job creation 创造就业 (chuàngzào jiùyè), job displacement 岗位替代 (gǎngwèi tìdài)

Activity 1 (15 min): "Then and Now" — Compare shopping, banking, communication in 1990 vs. 2024. What changed?

Activity 2 (15 min): Debate — "ICT creates more jobs than it destroys." Arguments for and against.

Activity 3 (10 min): Digital divide research — What percentage of the world has internet access? Why does this matter for education?

Exam Tip: "Discuss" questions require balanced arguments. Always present BOTH sides: "On one hand... On the other hand..." Conclude with a justified opinion.

Week 13 — Effects of ICT: Environmental & Health

Syllabus: 6.3–6.4 Environmental and Health Effects

Keywords: e-waste 电子垃圾 (diànzǐ lājī), recycling 回收利用 (huíshōu liyòng), carbon footprint 碳足迹 (tàn zújì), data center 数据中心 (shùjù zhōngxīn), RSI 重复性劳损 (chóngfù xìng sǔnshāng), eye strain 眼睛疲劳 (yǎnjīng pí láo), ergonomics 人体工程学 (rén tǐ gōngchéng xué)

Activity 1 (15 min): E-waste calculation — How many old phones are in your household? Class total? Research recycling options in Shenzhen.

Activity 2 (15 min): Ergonomics audit — Check your workstation: chair height, monitor distance, wrist position, lighting.

Activity 3 (10 min): "The Hidden Cost of Cloud" — Data centers use 1% of global electricity. Discuss sustainable solutions.

Week 14 — Safety & Security: Threats

Syllabus: 9.1 Security Threats

Keywords: hacking 黑客攻击 (hēikè gōngjī), malware 恶意软件 (èyì ruǎnjiàn), virus 病毒 (bīngdú), worm 蠕虫 (rúchóng), trojan 木马 (mù mǎ), spyware 间谍软件 (jiàndié ruǎnjiàn), adware 广告软件 (guǎnggào ruǎnjiàn), ransomware 勒索软件 (lèsuǒ ruǎnjiàn), phishing 网络钓鱼 (wǎngluò diàoyú), pharming 域欺骗 (yù qīpiàn)

Activity 1 (15 min): Malware matching — Match type to description, example, and damage caused.

Activity 2 (15 min): Case study analysis — WannaCry ransomware attack (2017). How did it spread? What was the impact? How could it have been prevented?

Activity 3 (10 min): "Phishing vs. Pharming" — Compare: phishing tricks the user; pharming redirects traffic without user knowledge.

Week 15 — Safety & Security: Countermeasures

Syllabus: 9.2–9.3 Security Measures

Keywords: firewall 防火墙 (fánghuǒqiáng), antivirus 杀毒软件 (shādú ruǎnjiàn), encryption 加密 (jiāmì), SSL 安全套接层 (ānquán tàojiē céng), biometric 生物识别 (shēngwù shíbié), two-factor authentication 双因素认证 (shuāng yīn sù rènzhèng), backup 备份 (bèifèn), UPS 不间断电源 (bù jiànduàn diànyuán), access rights 访问权限 (fǎngwèn quánxiàn)

Activity 1 (15 min): "Defense in Depth" — Layered security diagram: firewall (outer) → antivirus (middle) → encryption (inner) → backup (recovery).

Activity 2 (15 min): Backup strategies — Compare full, incremental, differential backup. 3-2-1 rule: 3 copies, 2 media, 1 offsite.

Activity 3 (10 min): Biometrics debate — Fingerprint vs. facial recognition. Pros, cons, privacy concerns.

Exam Tip: For "how to protect against X" questions, use the layered approach: 1) Prevention (firewall, antivirus), 2) Detection (monitoring, logs), 3) Recovery (backup, UPS). This structure guarantees full marks.

Week 16 — Systems Analysis: The SDLC

Syllabus: 8.1 – 8.2 Systems Development Life Cycle

Keywords: SDLC 系统开发生命周期 (xìtǒng kāifā shēngmìng zhōuqī), analysis 分析 (fēnxī), design 设计 (shèjì), implementation 实施 (shíshī), testing 测试 (cèshì), evaluation 评估 (pínggū), maintenance 维护 (wéihù), feasibility study 可行性研究 (kěxíng xìng yánjiū)

Activity 1 (15 min): SDLC timeline — Students create a visual timeline of the 6 stages with key activities in each.

Activity 2 (15 min): Case study — A school wants a new library system. What happens in each SDLC stage?

Activity 3 (10 min): Feasibility types — Technical, economic, operational, legal. Which is most important?

Week 17 — Systems Analysis: Methods & Documentation

Syllabus: 8.3–8.5 Analysis Methods and Documentation

Keywords: data flow diagram (DFD) 数据流图 (shùjù liú tú), system flowchart 系统流程图 (xìtǒng liúchéng tú), IPO chart 输入处理输出图 (shūrù chǔlǐ shūchū tú), pseudocode 伪代码 (wěi dàimǎ), algorithm 算法 (suànfǎ), entity 实体 (shí tǐ), process 处理 (chǔlǐ), data store 数据存储 (shùjù cúnchǔ)

Activity 1 (15 min): DFD drawing — Draw a Level 0 DFD for an online shopping system: Customer, Website, Order Database, Payment Processor.

Activity 2 (15 min): Pseudocode writing — Write pseudocode for "calculate total price including tax."

Activity 3 (10 min): Flowchart symbols — Match: oval (start/end), rectangle (process), diamond (decision), parallelogram (input/output).

Exam Tip: DFD questions often ask you to identify missing entities or data flows. Check: Is there a source for every input? A destination for every output? Is the data store read from AND written to?

Week 18 — Semester 1 Revision & Assessment

Mock Paper 1 (40 min):

Students complete a shortened Paper 1 covering all Semester 1 topics:

Define: LAN, WAN, protocol, firewall [4 marks]

Describe the fetch-decode-execute cycle [4 marks]

Compare SSD and HDD for a gaming laptop [3 marks]

Explain how DNS works when accessing a website [3 marks]

Discuss the social effects of increased ICT use in education [6 marks]

Revision Strategy: Create topic cards with: 1) Key definition, 2) Diagram, 3) Example, 4) Past question. Review 5 cards per day.

Standards & Syllabus Mapping

Weeks	Syllabus	Topic	Past Paper Focus
10 – 11	5.1 – 5.3	Data Types & Structures	MJ 2023 Q2, ON 2022 Q4
12 – 13	6.1 – 6.4	Effects of ICT	MJ 2022 Q6, ON 2021 Q7
14 – 15	9.1 – 9.3	Safety & Security	MJ 2023 Q5, ON 2022 Q6
16 – 17	8.1 – 8.5	Systems Analysis	MJ 2022 Q8, ON 2021 Q9